Musicology - Structural Correspondence with the 5-Stage Model

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Musicology

Structural correspondence survey with the 5-stage model (Ba / Wave / En / Uzu / Taba)

Survey Overview

- **Survey target**: 7 major theories in musicology
- **Research question**: Do the various theories of musicology correspond structurally to the 5-stage model?
- **Results**: Strong correspondence in 1 case, partial correspondence in 1 case, conditional correspondence in 2 cases

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Overview of the 5-Stage Model

Stage	Definition
Ba (Field)	An undifferentiated state. The initial condition in which no direction or structure has yet been established
Wave	The exploratory stage in which multiple directions diverge and compete
En (Edge)	A state of tension in which opposing elements coexist without converging on either side. The place where they meet at the boundary, influence each other, and relationships emerge
Uzu (Vortex)	The stage in which a new coherence (order) arises spontaneously from within the tension
Taba (Bundle)	The stage in which the form is fixed and stabilizes as a reusable structure

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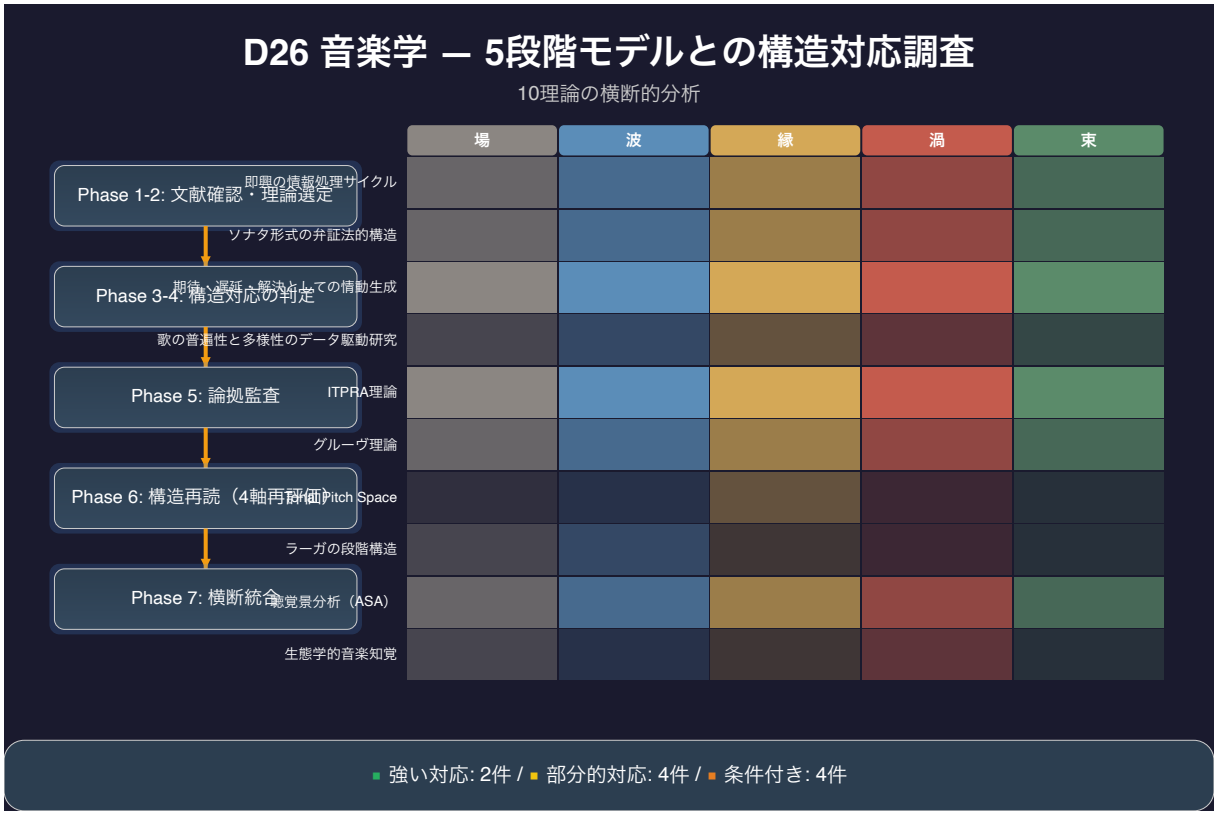

Figure 1: Domain — Research Overview Infographic

Figure 2: Domain — Theory x 5-Stage Correspondence Matrix

Overview of Structural Correspondences

Confidence	Theory group	Positioning
Near-established	Meyer's expectation theory, Huron's ITPRA theory	The process of "expectation→deviation→re-evaluation" corresponds with high precision to all 5 stages
Compelling	Pressing's improvisation model, Sonata form, Groove theory, Auditory scene analysis (Bregman), Universality of song research (NHS)	Correspondence exists, but differences in level of description (information processing vs. ontological generation) and distinction from the meta-level are needed
Conditional	Lerdahl TPS, Raga's stage structure, Clarke's ecological music perception	Partial correspondence exists, but with caveats regarding the analytical model character, limited textual basis, and theoretical critique of stage models

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Key Entry 1: Musical Expectation and Emotion Generation (Meyer)

- Meyer (1956) proposed a theory in which musical meaning arises from the generation, inhibition, delay, and resolution of expectations. Listeners learn musical tendencies within a specific style system, and when those tendencies are inhibited or delayed, suspense is created and alternative continuations arise. After unexpected events, listeners rapidly try to assimilate them into their belief system. This theory describes the mechanism of musical emotion generation as a 5-stage process of "expectation → inhibition → alternative possibility → resolution → belief integration."
- **As a matter of fact**: Meyer argued that musical meaning and emotion arise from the generation, inhibition, delay, and resolution of expectations (tendencies) learned within a specific style system. When a tendency is inhibited, suspense is produced, and emotional tension is maximized in the scene where alternative continuations stand in parallel. The correspondence between expectation violation and physiological responses (galvanic skin response, heart rate, etc.) has also been confirmed in subsequent experimental research (Egermann et al., 2013).
- **As an interpretation of structure**: Here we read as a structural feature the state in which multiple continuations compete while it remains undetermined which will be realized. The level of similarity is process. Particular attention is given to the structure of "the parallelism and weighting of alternative continuations." The temporal unfolding — expectation arising, expectation being inhibited, multiple possibilities coexisting, outcome arriving, and integration into the belief system — corresponds to the 5 stages as a process.
- **As an interpretive reading**: The operation of cultural tendencies (style system) corresponds to Ba. The arising of expectation and its inhibition correspond to Wave. The phase in which alternative continuations stand in parallel and are weighted corresponds to En as a state of tension in which opposing elements coexist without converging on either side. Re-evaluation of the arriving outcome corresponds to Uzu, and integration into the belief system corresponds to Taba. This correspondence holds with high precision across all stages, making it the core connection point of the 5 stages in musicology.

Key Entry 2: ITPRA Theory (Huron)

- Huron (2006) reconstructed Meyer's expectation-emotion theory within the framework of evolutionary psychology and computational neuroscience. Five functionally independent response systems are said to operate in temporal order in response to musical expectation: Imagination (anticipatory imaginative rehearsal), Tension (anticipatory physical preparation), Prediction (reward for prediction

accuracy at the moment of outcome), Reaction (automatic response at the moment of outcome), and Appraisal (post-hoc conscious evaluation).

- **As a matter of fact**: Huron discusses that the Prediction response is an independent mechanism that generates positive emotion in response to prediction accuracy, and the structure of "contrastive valence," in which the negative emotion of the Tension response converts to positive emotion at resolution. This provides an evolutionary explanation for why "relief after anxiety" is pleasurable.
- **As an interpretation of structure**: Here we read the sequential structure in which 5 response systems unfold temporally. The level of similarity is structure, and it is not merely that labels are similar, but that the functional content of each response system and the definition of each stage of the 5 stages correspond individually. In particular, the Prediction response describes computationally the very moment when the space of possibilities closes — the boundary moment when prediction and reality meet.
- **As an interpretive reading**: Imagination corresponds to Ba as imaginative rehearsal drifting in the future space of possibilities. Tension corresponds to Wave as arousal/generation of opposition through physical preparation. Prediction corresponds to En as the boundary moment when prediction and reality meet. Reaction corresponds to Uzu as the crystallization of automatic response. Appraisal corresponds to Taba as conscious re-evaluation and integration into the belief system. ITPRA theory answers "how do expectation response systems operate" to the question "why does music generate emotion" that Meyer asked, providing illumination from different levels of description of the same phenomenon.

Key Entry 3: Information Processing Cycle of Improvisation (Pressing)

- Pressing (1988) described improvisation as a 7-step cycle of "event generation → sensory monitoring → cognitive representation → evaluation → next action planning → next action → feedback." A theory at the intersection of improvisation research and cognitive science, it is suggested that the difference between improvisation and fixed performance is greatest at the stages of representational evaluation and next-action design.
- **As a matter of fact**: In Pressing's model, errors in improvisation are not targets for correction but become inputs for the next creation. It is also suggested that improvisational skill depends on the richness of schemas (the thickness of accumulated layers of musical patterns). This cyclic structure operates simultaneously across multiple scales: milliseconds (motor control), seconds (phrase construction), and years (style formation).
- **As an interpretation of structure**: Here we read the structure in which "cognitive representation and evaluation" function as a qualitative turning point within a cyclic feedback structure. The level of similarity is process. Attention is given to the point that the 7-step cycle is not mere repetition but that a new musical context is generated with each cycle.
- **As an interpretive reading**: The continuous background prior to action (musical context and style knowledge) corresponds to Ba. Difference detection through sensory monitoring corresponds to Wave. Cognitive representation and evaluation — the boundary where sound becomes music — correspond to En. The design and firing of the next action sequence correspond to Uzu. And the accumulation of cyclic repetition and the formation of style correspond to Taba, with the cyclic structure in which that Taba defines the next Ba. It should be noted, however, that Pressing's model is an information-processing description, which is at a different level from an ontological-generative description.

Cross-Cutting Patterns

- The following structural patterns recur across musicology
- **First, the recursive structure of "expectation → deviation → re-evaluation"
- **Second, the multilayered description of En
- **Third, the cyclic structure of Taba → Ba

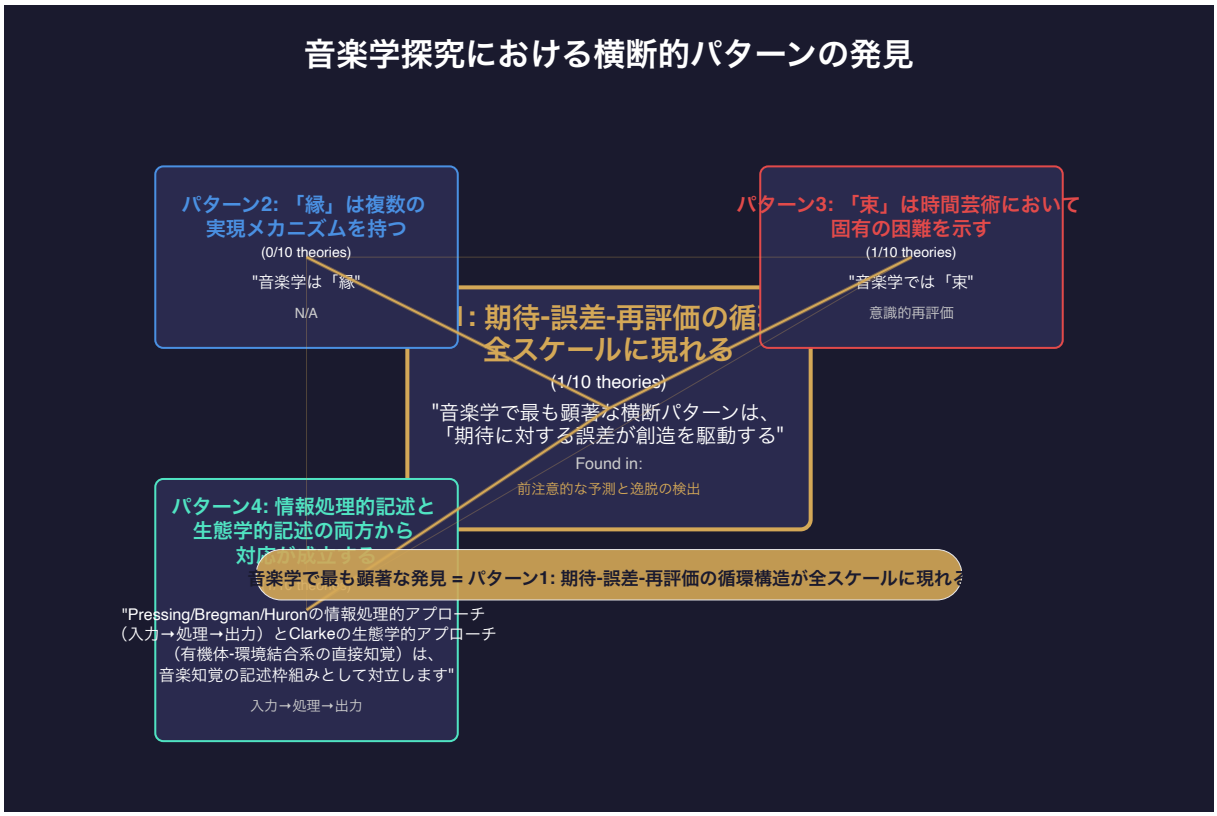

Figure 3: Domain — Cross-Cutting Patterns Diagram

Unresolved Questions

- The relationship between Meyer's expectation theory and predictive coding has not yet been sufficiently sorted out. Whether Meyer's (1956) "tendency" is identical to the computational "prediction" or is a different concept is an unresolved question spanning both music psychology and cognitive science.
- How to describe the relationship between "the subject designing opposition" and "form developing autonomously" in sonata form is a theoretical problem relating to the non-dual relationship between composition and form.
- Where to draw the line between analytical models like Lerdahl TPS and generative descriptions like Pressing's improvisation model remains unresolved. "Analysis" and "generation" may be located on a spectrum, or it may be an error to draw the line at all.
- Whether the transitional zone from jor to jhala in raga corresponds to En or is closer to "amplification of Wave" is a question relating to the interpretation of stage structures in non-Western music.

Conclusion

- This survey confirmed that musicology is a domain with overall strong structural similarity to the 5-stage model
- There are three particularly important findings